

2.0.3 Example:

Find the length of the curve $x = e^t + e^{-t}$, $y = 5 - 2t$, $0 \leq t \leq 3$

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2 &= (e^t - e^{-t})^2 = e^{2t} - 2e^t e^{-t} + e^{-2t} \\ &= e^{2t} - 2 + e^{-2t}\end{aligned}\qquad \left(\frac{dy}{dt}\right)^2 = (-2)^2 = 4$$

$$dS = \sqrt{e^{2t} - 2 + e^{-2t} + 4} dt = \sqrt{e^{2t} + 2 + e^{-2t}} dt$$

$$= \sqrt{(e^t + e^{-t})^2} dt = (e^t + e^{-t}) dt$$

$$\begin{aligned}\int_0^3 e^t + e^{-t} dt &= \left[e^t - e^{-t} \right]_0^3 = \left(e^3 - \frac{1}{e^3} \right) - (1 - 1) \\ &= \boxed{e^3 - \frac{1}{e^3}}\end{aligned}$$

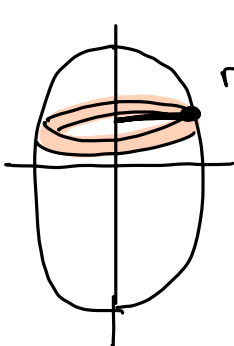
3 Parametric Surface Area of Revolution

Recall: Surface area $= 2\pi \int_{\alpha}^{\beta} (\text{radius}) dS$

Here we use the parametric dS , and we have to be sure that it's only traversed once.

3.0.1 Example:

The curve $x = 2 \sin t$, $y = 3 \cos t$, $0 \leq t \leq \pi$ is rotated around the y axis. Find the surface area.

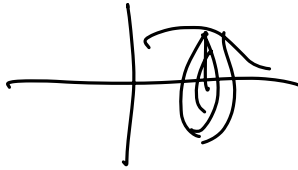

$$\begin{aligned}\text{radius} = x \quad dS &= \sqrt{(2 \cos t)^2 + (-3 \sin t)^2} dt \\ &= \sqrt{4 \cos^2 t + 9 \sin^2 t} dt \\ \text{S.A.} &= 2\pi \int_0^{\pi} 2 \sin t \sqrt{4 \cos^2 t + 9 \sin^2 t} dt\end{aligned}$$

3.0.2 Example:

9) The curve $x = 3t - t^3$, $y = 3t^2$, $0 \leq t \leq 1$ is rotated around the x -axis. Find the surface area.

$$\text{radius} = y = 3t^2$$

$$dS = \sqrt{(3-3t^2)^2 + (6t)^2} dt$$



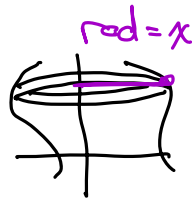
$$= \sqrt{9t^4 - 18t^2 + 9 + 36t^2} dt$$

$$= \sqrt{9t^4 + 18t^2 + 9} dt = \sqrt{(3t^2 + 3)^2} dt$$

$$dS = (3t^2 + 3) dt$$

$$S.A. = 2\pi \int_0^1 3t^2 (3t^2 + 3) dt$$

b) Rotated around y -axis: Now the radius is x
 $\text{rad} = 3t^2 - t^3$



$$S.A. = 2\pi \int_0^1 \underbrace{(3t^2 - t^3)}_{\text{radius}} \underbrace{(3t^2 + 3)}_{dS} dt$$

Wednesday

1. HW 8.2 and 10.1 Due Tonight
 2. Finish 10.2
 - A. Worksheet
 3. Start 10.3 Polar coordinates
-

$$x = \sqrt{4 - y^2}$$

$$0 \leq y \leq 1 \quad \text{in } y\text{-axis}$$

$$\text{radius} = x$$

$$\frac{dx}{dy} = \frac{-2y}{2\sqrt{4-y^2}} = \frac{-y}{\sqrt{4-y^2}}$$

$$dS = \sqrt{1 + \left(\frac{y^2}{4-y^2}\right)} dy$$

$$= \sqrt{\frac{4-y^2}{4-y^2} + \frac{y^2}{4-y^2}} dy$$

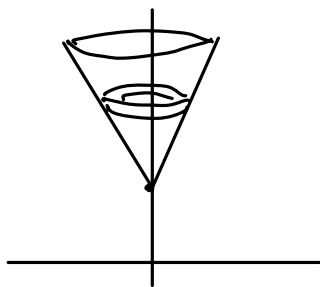
$$= \sqrt{\frac{4}{4-y^2}} dy = \boxed{\frac{2}{\sqrt{4-y^2}} dy = dS}$$

$$S.A. = 2\pi \int_0^1 \cancel{\sqrt{4-y^2}} \cdot \frac{2}{\cancel{\sqrt{4-y^2}}} dy$$

$$= 2\pi \int_0^1 2 dy = 4\pi y \Big|_0^1 = 4\pi(1-0) = 4\pi$$

8.2 (5)

$$y = \underline{6x+6} \quad 0 \leq x \leq 2 \quad \textcircled{2} \quad y\text{-axis}$$



$$dS = \sqrt{1 + (6)^2} = \sqrt{37} dx$$

$$\text{radius} = x$$

$$2\pi \int_0^2 \underline{x} \sqrt{37} dx = \pi 2\sqrt{37} \left[\frac{x^2}{2} \right]_0^2$$

$$= \sqrt{37} (4 - 0) \pi$$

$$= 4\sqrt{37} \pi$$